



PUSL3190 Computing Project

Abstract

Intellimeet

Supervisor: Ms. Sanuli Weerasinghe

Name: Parana Sasindi

Plymouth Index Number:10953782

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With the rapid digital transformation of higher education, efficient communication between students and academic staff is critical. However, many institutions still rely on informal scheduling methods like email and messaging apps, which lack organization and transparency. At NSBM Green University, this leads to significant challenges, including frequent scheduling conflicts, long wait times for meeting approvals, and a lack of visibility into lecturer availability. These inefficiencies directly delay project progress and increase the administrative burden on faculty.

To address these issues, this project introduces IntelliMeet, a mobile application designed to centralize and automate academic meeting coordination. Unlike generic scheduling tools, IntelliMeet is tailored for the academic environment, featuring role-based access for students and Lecturers . The system uses a client-server architecture with separate interfaces for students and lecturers. It applies JWT-based authentication for secure access. The system includes an AI feedback feature that reviews project abstracts and gives mock scores. It also provides a chatbot for real-time help. The backend enforces strict validation rules and prevents double booking in schedules.

The development follows an Agile methodology, allowing for iterative improvements based on continuous feedback. The system is built using a modern JavaScript and TypeScript stack. The frontend uses React Native and Expo for cross-platform mobile development. The backend uses Node.js with Express.js to provide a scalable architecture. MongoDB Atlas is used for flexible data management with Mongoose for object modeling. Security is handled using Bcrypt. The system is validated through automated API workflows that simulate full user lifecycles, including authentication and concurrency testing to prevent double booking. Snapshot testing is used on the frontend to prevent visual regressions in UI components.

The system is tested using automated API scripts that simulate full user workflows. It checks authentication, role-based access, and scheduling logic. Concurrency testing ensures that schedule conflicts are blocked with proper error responses. The AI features are tested for stable responses and connectivity. The system improves and simplifies academic supervision processes